

Robotics 2

Impedance Control

Prof. Alessandro De Luca

Dipartimento di Ingegneria Informatica
Automatica e Gestionale Antonio Ruberti



Sapienza
Università di Roma

Impedance control



- imposes a desired dynamic behavior to the interaction between robot end-effector and environment (in Cartesian space)
- the desired behavior is specified through an impedance model, namely a complete set of mass-spring-damper equations (typically chosen as linear and decoupled, but also nonlinear)
- a description of how a reaction wrench (= force/moment) is generated in response to environment deformation is not required
- suited for robot tasks in which the contact wrench should be "kept small", while its accurate regulation is not mandatory
- a control loop driven by the error on the contact wrench is missing, thus it is only indirectly assigned by controlling end-effector pose
- choosing the stiffness in the impedance model is a trade-off between contact wrench and pose accuracy (along each Cartesian direction)
- basic scheme in Cartesian space can be generalized to any task space

Dynamic model of a robot in contact

$q \in \mathbb{R}^n$ $\quad M(q)\ddot{q} + c(q,\dot{q}) + g(q) = \tau + \tau_{\text{int}}$ $\quad$ interaction with the environment

- duality between end-effector twists and wrenches $t = \begin{pmatrix} v \\ \omega \end{pmatrix} \in \mathbb{R}^6$ $w = \begin{pmatrix} f \\ \mu \end{pmatrix} \in \mathbb{R}^6$

$$t = J(q)\dot{q} \;\Rightarrow\; w^T t = w^T J(q)\dot{q} = \tau^T \dot{q} \;\Rightarrow\; \tau = J^T(q)w$$

geometric Jacobian $6 \times n$ — principle of virtual work

$$\Rightarrow \quad \tau_{\text{int}} = J^T(q)w$$

- with a minimal representation of the end-effector pose

$$y_e = \begin{pmatrix} p \\ \phi \end{pmatrix} = k_e(q) \in \mathbb{R}^6 \;\Rightarrow\; \dot{y}_e = \begin{pmatrix} \dot{p} \\ \dot{\phi} \end{pmatrix} = \frac{\partial k_e}{\partial q}\dot{q} = J_e(q)\dot{q}$$

analytic Jacobian $6 \times n$

$$t = \begin{pmatrix} v \\ \omega \end{pmatrix} = \begin{pmatrix} \dot{p} \\ \bar{T}(\phi)\dot{\phi} \end{pmatrix} = T(\phi)J_e(q)\dot{q} = J(q)\dot{q} \;\Rightarrow\; \tau_{\text{int}} = J_e^T(q)\underbrace{T^T(\phi)w}_{F_e \in \mathbb{R}^6} = J_e^T(q)F_e$$

- generic contact task (of any dimension $m \le n$)

$$y = k(q) \in \mathbb{R}^m \qquad J_y(q) = \frac{\partial k}{\partial q} \;\Rightarrow\; \tau_{\text{int}} = J_y^T(q)F_y$$

analytic task Jacobian $m \times n$

generalized contact force $F_y \in \mathbb{R}^m$ performing work on y

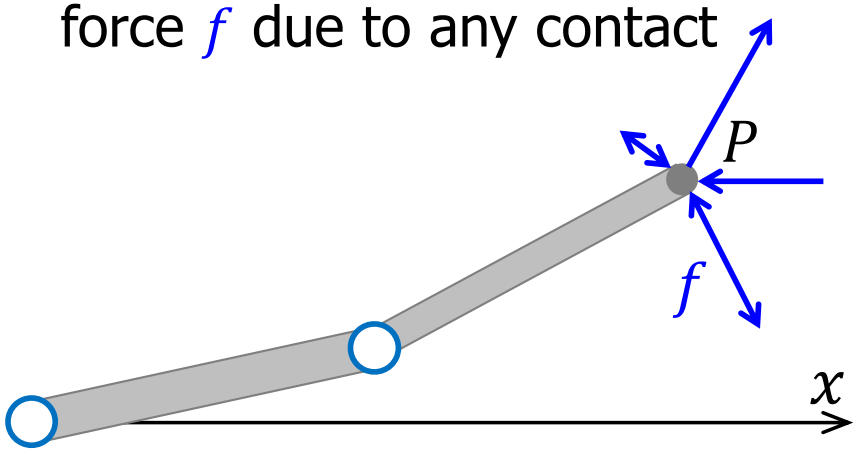
also denoted as $\gamma_{\text{int}} = -\gamma_{\text{env}} \in \mathbb{R}^m$ in new textbook

Contact forces vs. constraint forces

whiteboard…

$M(q)\ddot{q} + \cdots = \tau + J^T(q)f$ e.g., in $\mathbb{R}^2$ $M(q)\ddot{q} + \cdots = \tau + A^T(q)\lambda$

every possible direction of force f due to any contact

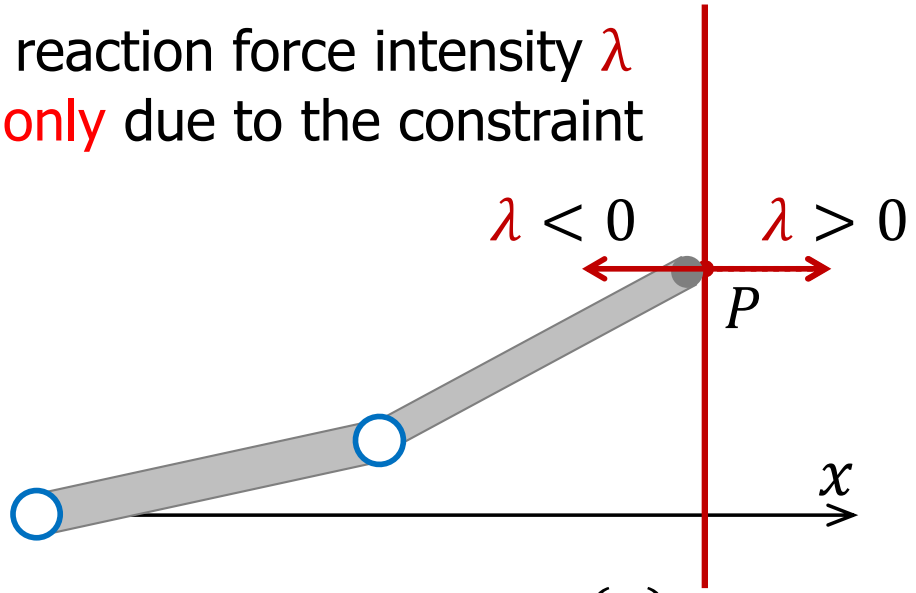


$$y = \begin{pmatrix} p_x(q) \\ p_y(q) \end{pmatrix} = k(q)$$

$$\dot{y} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \frac{\partial k}{\partial q}\dot{q} = J(q)\dot{q}$$

$$\Rightarrow f = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$$

reaction force intensity λ only due to the constraint



$$p_x(q) = c$$

$$\dot{p}_x = v_x = (1 \quad 0)\, J(q)\dot{q} = A(q)\dot{q} = 0$$

$$A^T(q)\lambda = J^T(q)\begin{pmatrix} 1 \\ 0 \end{pmatrix}\lambda = J^T(q)\begin{pmatrix} -f_x \\ 0 \end{pmatrix}$$

e.g., in (gravity-balanced) static equilibrium

$$\Rightarrow \lambda = -f_x$$

Dynamic model in task coordinates



assuming $m = n$, and out of singularities

$$M_y(q)\ddot{y} + c_y(q,\dot{q}) + g_y(q) = J_y^{-T}(q)\tau + F_y$$

$\tau_{\text{int}} = J_y^T(q)F_y$

with

$$M_y(q) = J_y^{-T}(q)M(q)J_y^{-1}(q) = \left(J_y(q)M^{-1}(q)J_y^T(q)\right)^{-1}$$

$$c_y(q,\dot{q}) = J_y^{-T}(q)\left(c(q,\dot{q}) - M(q)\dot{J}_y(q)J_y^{-1}(q)\dot{y}\right)$$

$$g_y(q) = J_y^{-T}(q)g(q)$$

task inertia matrix

... and the usual structural properties

- $M_y > 0$, when J_y is nonsingular
- $c_y(q,\dot{q}) = C_y(q,\dot{q})\dot{y}$
- $\dot{M}_y - 2C_y$ is skew-symmetric, if $\dot{M} - 2C$ satisfies the same property (e.g., with C given by Christoffel symbols)
- the task-space dynamic model can be linearly parametrized in terms of a set of dynamic coefficients

Design of the control law

designed in two steps:

1. feedback linearization in the task space (with force/torque measure)

$$\tau = J_y^T(q)\left(M_y(q)a + c_y(q,\dot{q}) + g_y(q) - F_y\right)$$

$$\ddot{y} = a$$ closed-loop system

2. imposition of a dynamic impedance model

$$M_m(\ddot{y} - \ddot{y}_d) + B_m(\dot{y} - \dot{y}_d) + K_m(y - y_d) = F_y$$

most often "decoupled" (diagonal matrices)

desired (apparent) inertia (> 0)

desired damping (> 0)

desired stiffness (> 0)

generalized contact forces from environment

which is realized by choosing

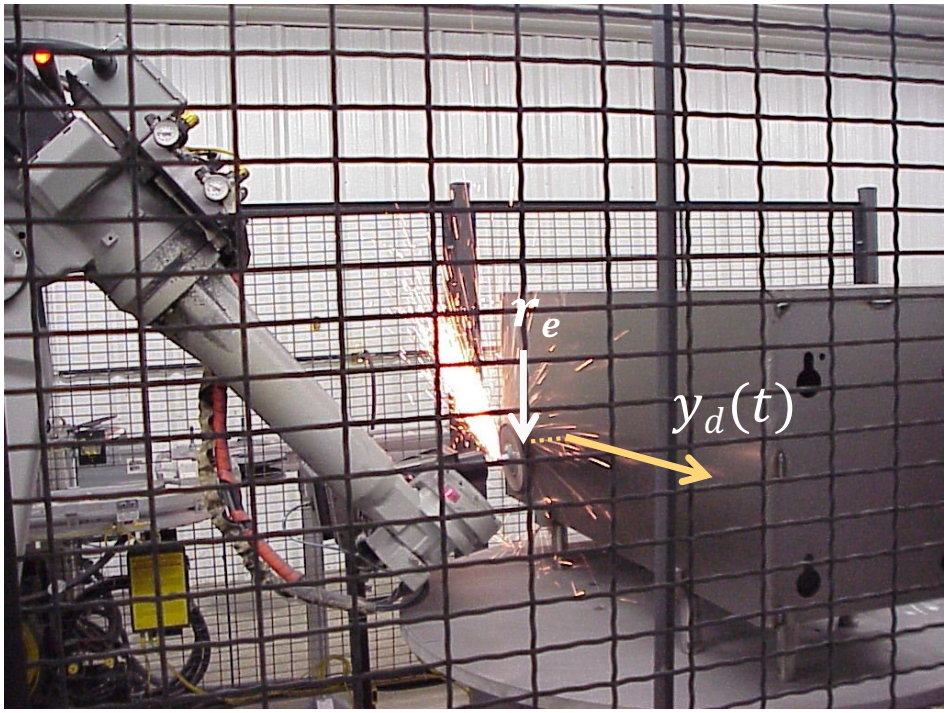
$$a = \ddot{y}_d + M_m^{-1}\left(B_m(\dot{y}_d - \dot{y}) + K_m(y_d - y) + F_y\right)$$

typically, the desired trajectory $y_d(t)$ "slightly penetrates" inside the compliant environment (inducing contact forces) ...

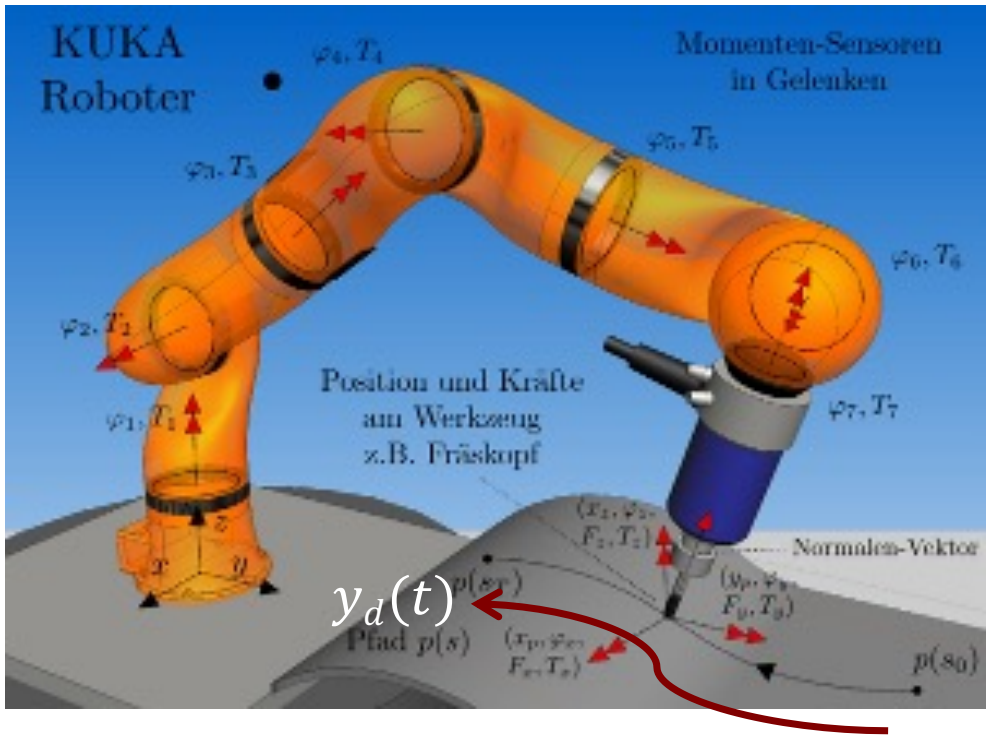
Examples of desired trajectory $y_d(t)$
in impedance/compliance control

$$M_m(\ddot{y} - \ddot{y}_d) + B_m(\dot{y} - \dot{y}_d) + K_m(y - y_d) = F_y$$

the desired motion $y_d(t)$ is slightly inside
the environment (thus enforcing contact)

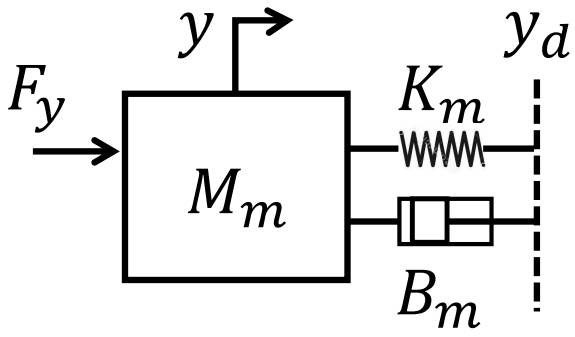


robot in grinding task



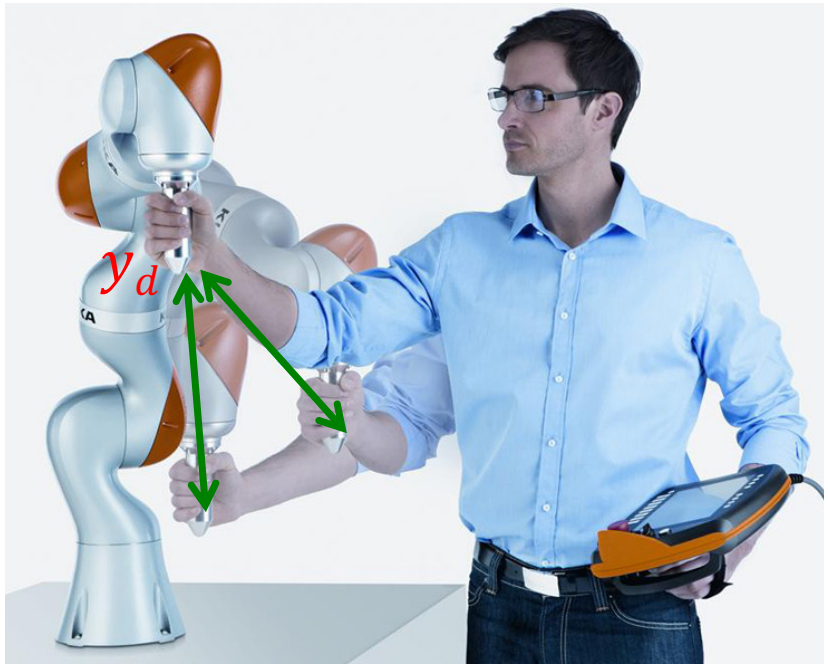
robot writing on a surface

Examples of desired reference y_d
in impedance control

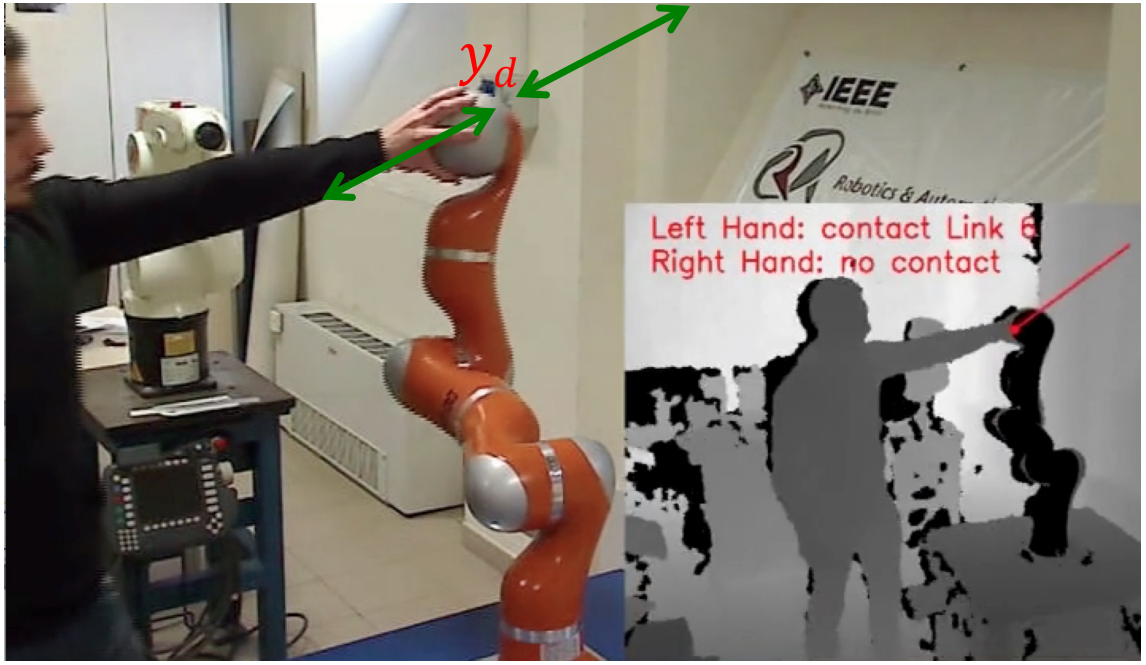


$$M_m(\ddot{y} - \ddot{y}_d) + B_m(\dot{y} - \dot{y}_d) + K_m(y - y_d) = F_y$$

constant desired pose y_d is the free Cartesian rest position in a human-robot interaction task



KUKA iiwa robot with human operator



KUKA LWR robot @DIAG in pHRI (collaboration)

Control law in joint coordinates



substituting and simplifying…

$$\tau = M(q)J_y^{-1}(q)\left(\ddot{y}_d - \dot{J}_y(q)\dot{q} + M_m^{-1}\left(B_m(\dot{y}_d - \dot{y}) + K_m(y_d - y)\right)\right) + c(q,\dot{q}) + g(q) + J_y^T(q)\left(M_y(q)M_m^{-1} - I\right)F_y$$

matrix weighting the measured contact forces

- the following identity holds for the term involving contact forces

$$J_y^T(q)\left(M_y(q)M_m^{-1} - I\right)F_y = \left(M(q)J_y^{-1}(q)M_m^{-1} - J_y^T(q)\right)F_y$$

  which eliminates from the control law also the presence of the last remaining dynamic task-based quantity (the inertia matrix $M_y(q)$)

- while the control design is based on dynamic analysis and desired (impedance) behavior described in the task/Cartesian space, the final control implementation is always at the robot joint level

Choice of the impedance model

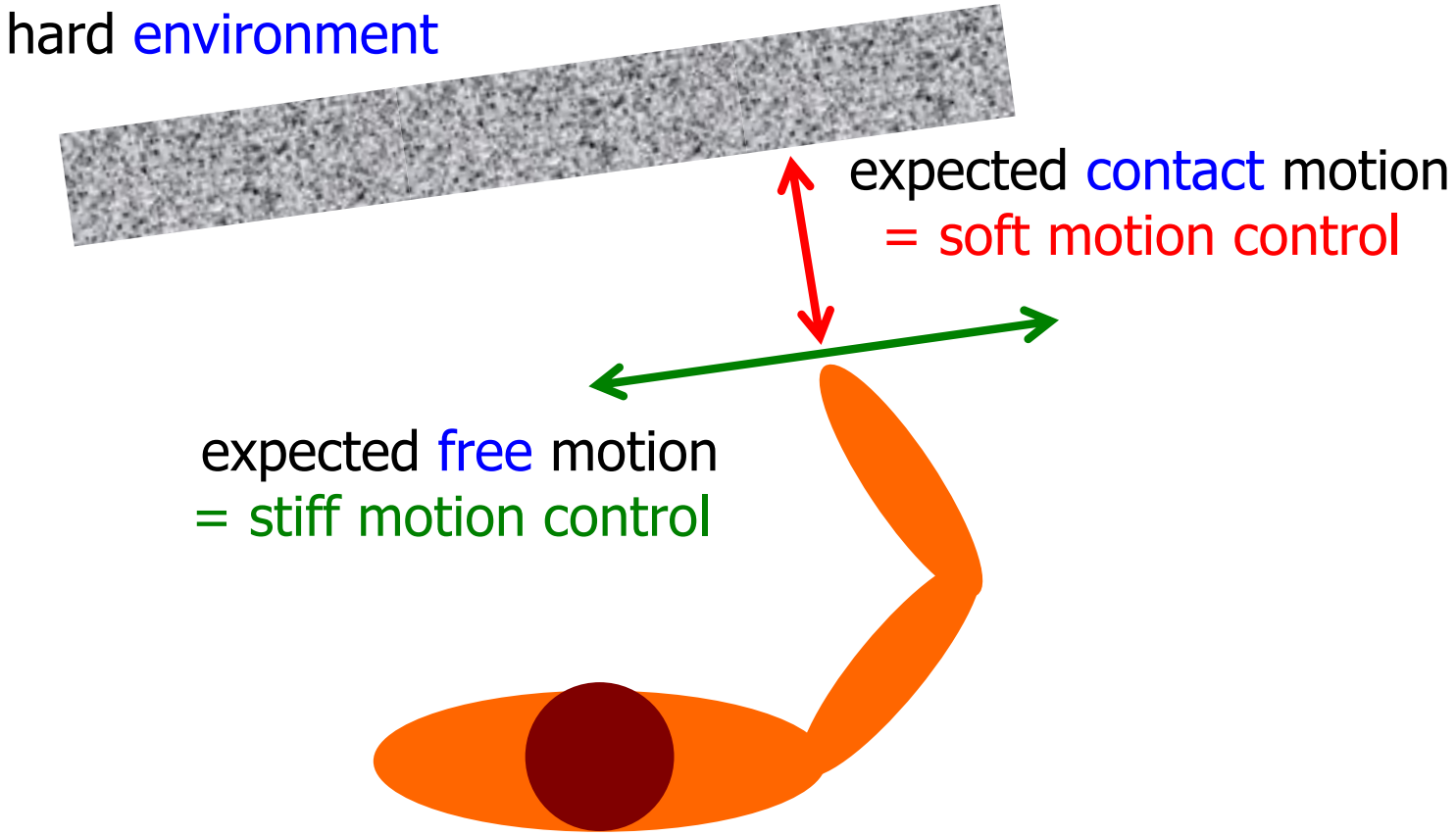


rationale ...

- adapt/match to the dynamic characteristics of the environment (in particular, of its estimated stiffness) in a complementary way
- avoid large impact forces due to uncertain geometric characteristics (position and/or orientation) of the environment
- mimic the behavior of a human arm
  - ➔ fast and stiff in "free" motion, slow and compliant in "guarded" motion

- large $M_{m,i}$ and small $K_{m,i}$ in task directions where a (hard) contact is foreseen (➔ low contact forces)
- large $K_{m,i}$ and small $M_{m,i}$ in task directions that are supposed to be free (➔ good tracking of desired motion trajectory)
- damping coefficients $B_{m,i}$ are then used to shape transient behaviors

Human arm behavior

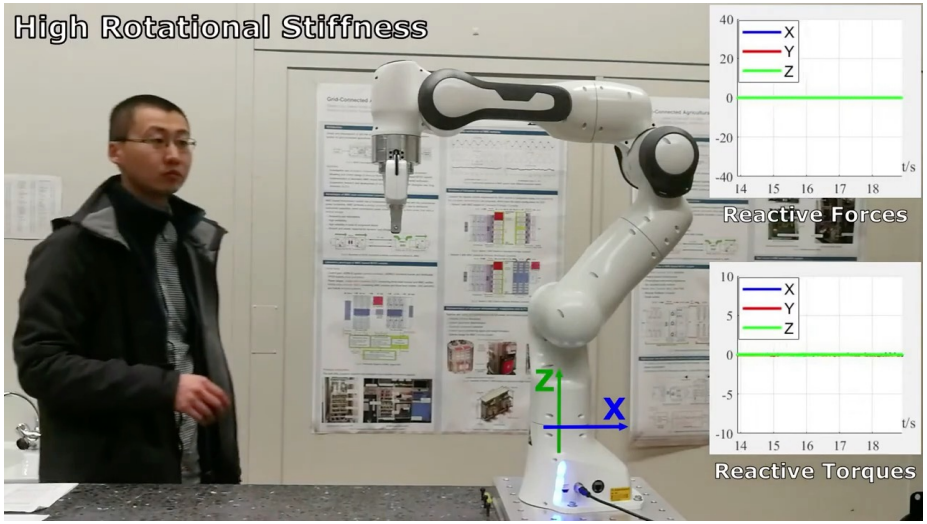


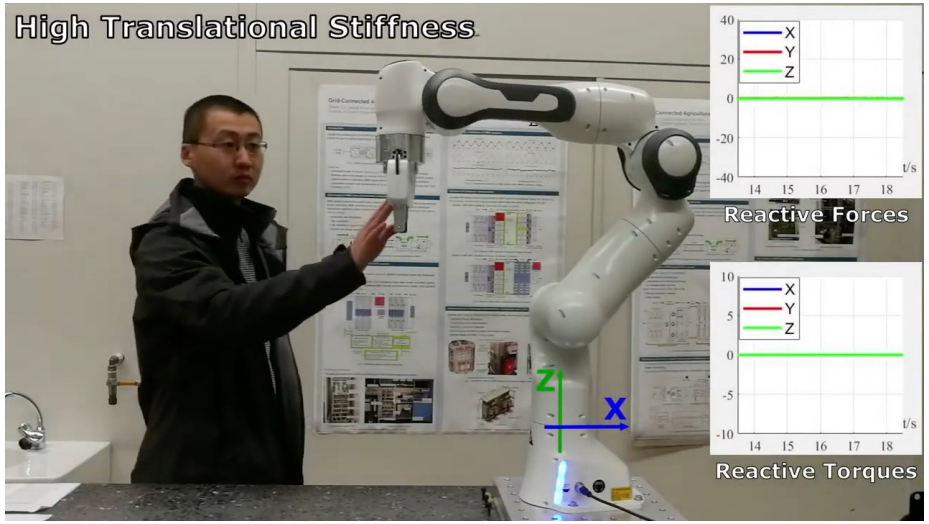
in the selected i-th task/Cartesian direction,
the stiffer is the environment, the softer should be the stiffness $K_{m,i}$ in model

Experiments with impedance control
human interaction with a Panda robot

7R Franka Emika Panda robot

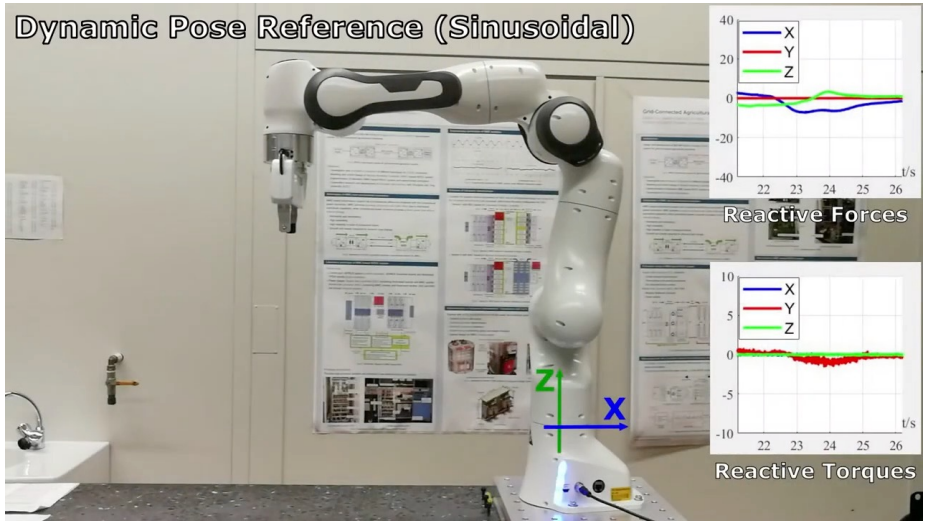
3 clips of the same video





high rotational $K_{m,\phi}$
low $K_{m,p}$
(compliant in position)

high translational $K_{m,p}$
low $K_{m,\phi}$
(compliant in orientation)



trajectory tracking with
physical interaction
(uniformly compliant)

LRS – RPTU
Kaiserlautern

A notable simplification - 1

choose the apparent inertia equal to the robot natural task/Cartesian inertia

$$M_m = M_y(q) = J_y^{-T}(q) M(q) J_y^{-1}(q) = \left(J_y(q)\, M^{-1}(q)\, J_y^T(q)\right)^{-1}$$

then, the control law becomes

$$u = M(q) J_y^{-1}(q)\left(\ddot{y}_d - \dot{J}_y(q)\dot{q}\}\right) + c(q,\dot{q}) + g(q) + J_y^T(q)\left(B_m(\dot{y}_d - \dot{y}) + K_m(y_d - y)\right)$$

WITHOUT contact force feedback! (a F/T sensor is no longer needed …)

this is a pure motion control law applied also during interaction,
but designed so to keep limited contact forces at the task/end-effector level
(as before, K_m is chosen as a function of the expected environment stiffness)

A notable simplification - 2

technical issue: if the impedance model (now, nonlinear) is still supposed to represent a real mechanical system, then in correspondence to a desired non-constant inertia ($M_y(q)$) there should be Coriolis and centrifugal terms ...

$$M_y(q)(\ddot{y} - \ddot{y}_d) + \left(C_y(q,\dot{q}) + B_m\right)(\dot{y} - \dot{y}_d) + K_m(y - y_d) = F_y$$

nonlinear impedance model ("only" gravity terms disappear)

redoing computations, the control law becomes

$$\tau = M(q)J_y^{-1}(q)\left(\ddot{r}_d - \dot{J}_y(q)J_y^{-1}(q)\dot{y}_d\right) + C(q,\dot{q})J_y^{-1}(q)\dot{y}_d + g(q) + J_y^T(q)\left(B_m(\dot{y}_d - \dot{y}) + K_m(y_d - y)\right)$$

which is slightly more complex, but has the following advantages

- guarantees asymptotic convergence to zero tracking error (on $y_d(t)$) when $F_y = 0$ (no contact situation) ⇒ Lyapunov + skew-symmetry of $\dot{M}_y - 2C_y$
- leads to further simplifications when y_d is constant

Task/Cartesian regulation revisited
(without contact, $F_y = 0$)



when y_d is constant ($\dot{y}_d = 0$, $\ddot{y}_d = 0$), from the previous expression we get the control law

$$\tau = g(q) + J_r^T(q)(K_m(y_d - y) - B_m\dot{y}) \quad (\bigstar)$$

task/Cartesian PD control with gravity cancellation …

when $F_y = 0$ (absence of contact), we already know that this control law ensures asymptotic stability of y_d, provided $J_y(q)$ has full rank

proof (alternative)

Lyapunov candidate $V_1 = \frac{1}{2}\dot{y}^T M_y(q)\dot{y} + \frac{1}{2}(y_d - y)^T K_m(y_d - y)$

$$\dot{V}_1 = \dot{y}^T M_y(q)\ddot{y} + \frac{1}{2}\dot{y}^T \dot{M}_y(q)\dot{y} - \dot{y}^T K_m(y_d - y) = \cdots = -\dot{y}^T B_m \dot{y} \leq 0$$

using skew-symmetry of $\dot{M}_y - 2C_y$ and $g_y = J_y^{-T} g$

Task/Cartesian regulation
(with contact, $F_y \neq 0$)



when $F_y \neq 0$, convergence to y_d is not assured
(it may not even be a closed-loop equilibrium …)

- for analysis, assume an elastic contact model for the environment

$$F_y = K_e(y_e - y)$$ with stiffness $K_e \geq 0$ and rest position y_e

- closed-loop system behavior

Lyapunov candidate

$$V_2 = \frac{1}{2}\dot{y}^T M_y(q)\dot{y} + \frac{1}{2}(y_d - y)^T K_m (y_d - y) + \frac{1}{2}(y_e - y)^T K_e (y_e - y)$$

$$= V_1 + \frac{1}{2}(y_e - y)^T K_e (y_e - y)$$

$$\dot{V}_2 = \dot{y}^T M_y(q)\ddot{y} + \frac{1}{2}\dot{y}^T \dot{M}_y(q)\dot{y} - \dot{y}^T K_m (y_d - y) - \dot{y}^T K_e (y_e - y)$$

$$= \cdots = -\dot{y}^T B_m \dot{y} + \dot{y}^T (F_y - K_e(y_e - y)) = -\dot{y}^T B_m \dot{y} \leq 0$$

Stability analysis (with $F_y \neq 0$)



when $\dot{y} = \ddot{y} = 0$, at a closed-loop system equilibrium it is

$$K_m(y_d - y) + K_e(y_e - y) = 0$$

which has the unique solution

$$y = (K_m + K_e)^{-1}(K_m y_d + K_e y_e) =: y_E$$

in fact, the Lyapunov candidate V_2 has its minimum in y_E (check it!)

LaSalle ➡ y_E is an asymptotically stable equilibrium ◀

$$y_E \approx \begin{cases} y_e \text{ for } K_e \gg K_m \text{ (rigid environment)} \\ y_d \text{ for } K_m \gg K_e \text{ (rigid controller)} \end{cases}$$

➡ at steady state the contact force is $F_y = K_e(y_e - y_E)$

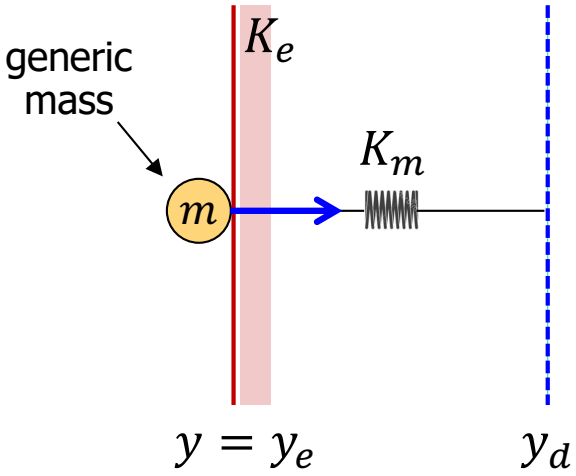
the law (★) is often called compliance (or stiffness) control in the literature

Equilibrium condition

whiteboard…

$$y_E = (K_m + K_e)^{-1}(K_m y_d + K_e y_e)$$

let $y \in \mathbb{R}$

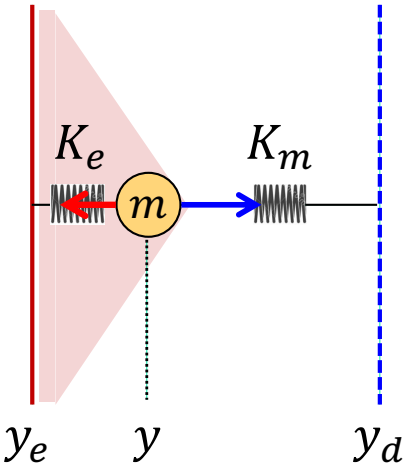


at the initial contact

$$m\ddot{y} = F_c - B_m\dot{y}$$

$$F_c = K_m(y_d - y_e) > 0$$

part of the task/Cartesian control force



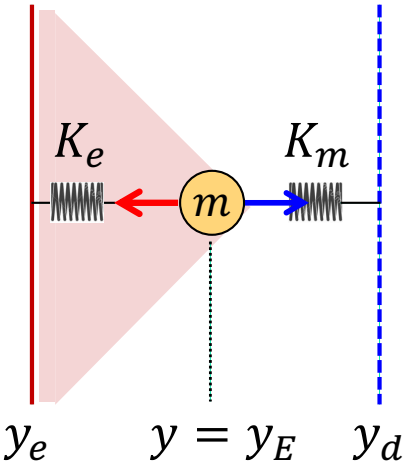
during the transient

$$m\ddot{y} = F_c + F_y - B_m\dot{y}$$

$$F_c = K_m(y_d - y) > 0$$

$$F_y = -K_e(y - y_e) < 0$$

$$|F_c| \neq |F_y|$$



at steady-state (equilibrium)

$$0 = F_c + F_y$$

$$F_c = K_m(y_d - y_E) > 0$$

$$F_y = -K_e(y_E - y_e) < 0$$

$$\Rightarrow \quad y_E = \frac{K_m y_d + K_e y_e}{K_m + K_e}$$

Active equivalent of a RCC device

IF

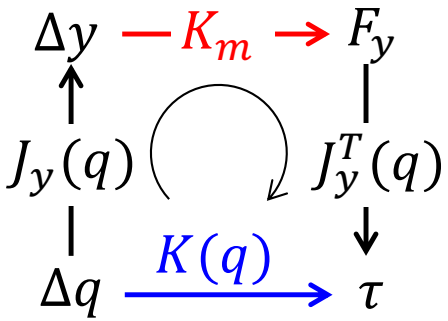
- displacements from the desired constant position y_d are small, namely

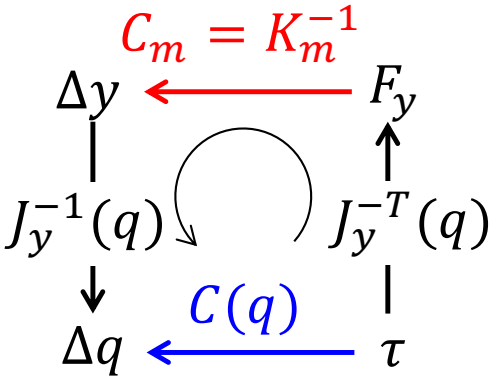
$$(y_d - y) \approx J_y(q)(q_d - q)$$

- $g(q) = 0$ (gravity compensated), $D_m = 0$ (or $\dot{y} \approx 0$, i.e., small enough)

THEN from (★)

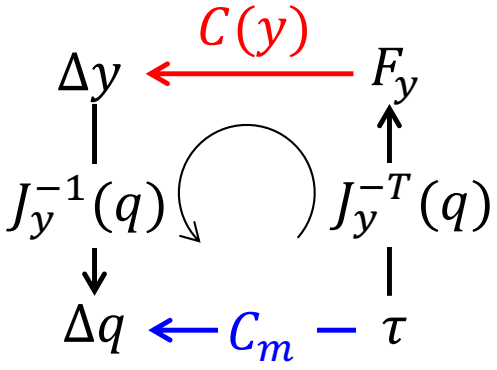
$$\tau = J_y^T(q) K_m J_y(q)(q_d - q) = K(q)(q_d - q)$$





constant Cartesian-level stiffness K_m
(or compliance $C_m = 1/K_m$)
corresponds to
variable joint-level stiffness $K(q)$
(or compliance $= C(q)$)

… and vice versa



this is the "active" counterpart of a
Remote Center of Compliance (RCC) device

Admittance control

- in some cases, we don't have access to the low-level robot torque (or motor current) commands ⇒ closed control architecture
- for handling the interaction with the environment, one uses often admittance control: contact forces F_c ⇒ velocity commands $\dot{q}$
- implementation (with compliant matrices C)
  - in joint space or in task (Cartesian) space – with singularity issues …
  - at the velocity or incremental position level

$$\tau_c = J^T(q)F_c \longrightarrow \dot{q} = C_q\tau_c \longrightarrow \boxed{\dot{q} = C_q\, J^T(q)F_c} \qquad C_q \geq 0$$

$$\updownarrow$$

Δq (to be added to the current q)

$$F_c \longrightarrow \dot{y} = C_y F_c \longrightarrow \boxed{\dot{q} = J^{-1}(q)C_y F_c} \qquad C_r \geq 0$$

$$\updownarrow$$

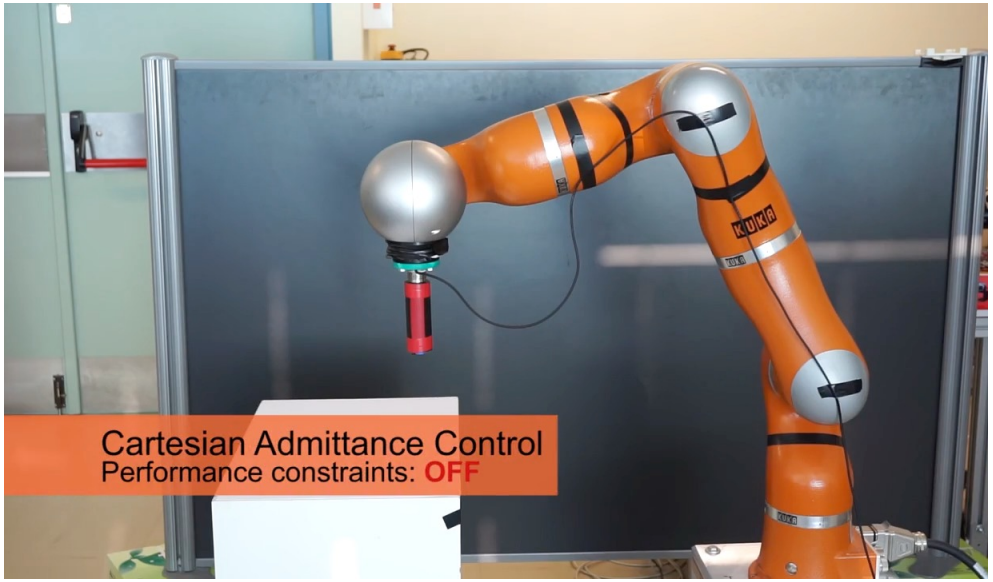
(in case of redundancy) $J^{\#}(q)$

Experiments with admittance control
human interaction with a KUKA LWR robot

7R KUKA LWR4+ robot

handling of task singularities
through performance constraints

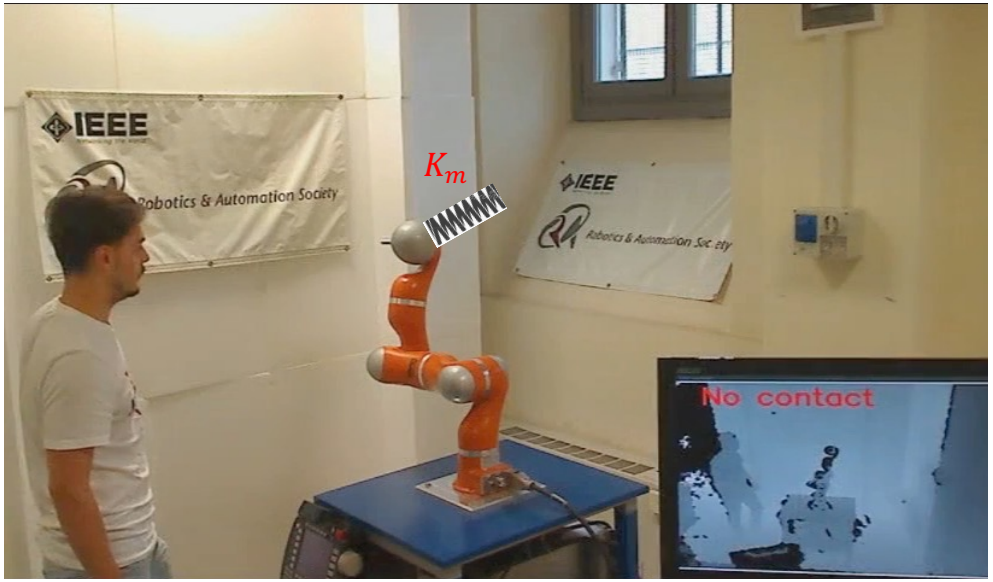
video



ICRA 2016, University of Patras

admittance control at any contact point
without using a force/torque sensor

video

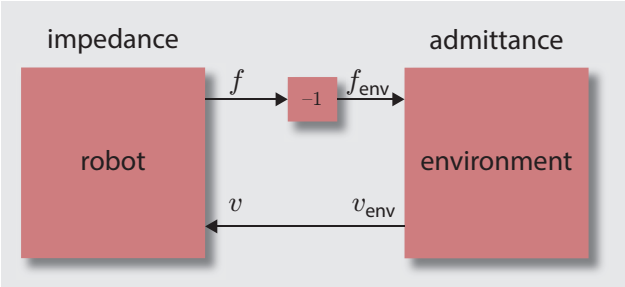


Sep 2013, DIAG Laboratory of Robotics

Impedance vs Admittance control

concept schematics of robot-environment interaction

a port-based approach



power exchange

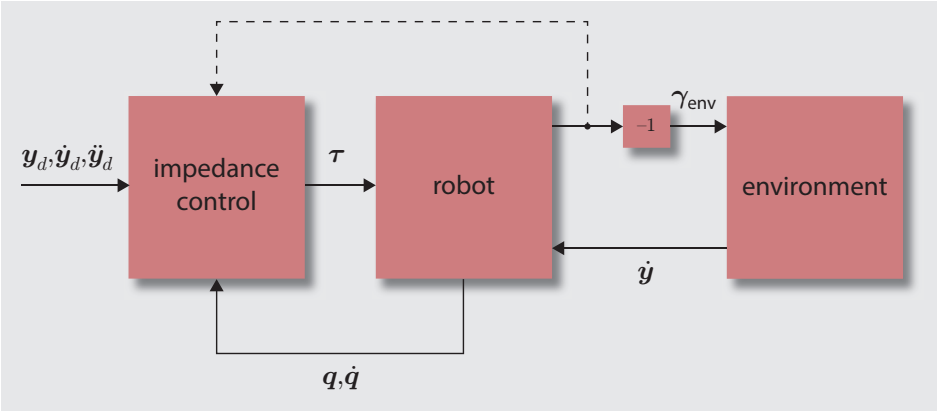
$$\mathcal{P} = f^T v$$

at an interaction port

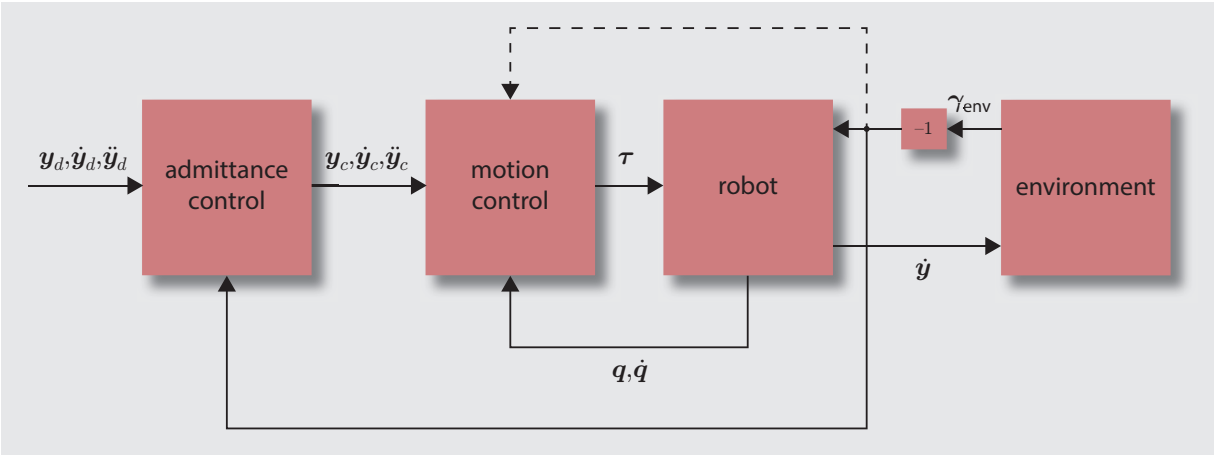
allows a conservative but powerful analysis based on the passivity concept

(e.g., passivity is preserved in series, parallel, or feedback connections of passive subsystems)

– a sufficient stability condition



generalized task impedance control



generalized task admittance control